

Discussion of randomization tests

Presentation Outline

1 Notes and thoughts on Randomization tests

- Exactly what is the null hypothesis?
- Why almost all tests are two-sided

2 Why aren't design-based tests used more frequently?

- Why do you usually see t-tests instead of randomization tests?
- Why randomization tests can be very useful

3 Important Points

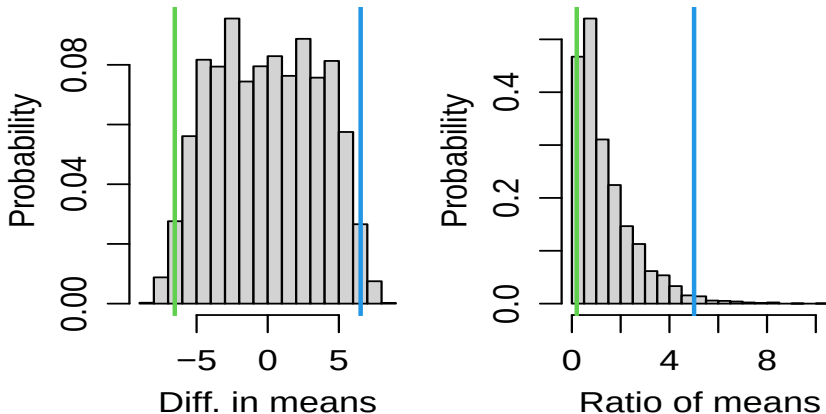
What exactly is the null hypothesis in my examples?

Goal: test whether two populations have the same mean

- But, usual randomization starts with all observations lumped together
- I.e., null hypothesis is that the two populations have the same distribution
 - More than equal means
- Generally ignored because the test statistic (difference of means, ratio of means) is most sensitive to unequal means

Why scientific practice is almost exclusively two-sided tests

Relationship between one- and two-sided p-values: Tomato study



Why scientific practice is almost exclusively two-sided tests

Relationship between one- and two-sided p-values: Tomato study

- Tomato: two-sided $p = 0.035$
 - One-sided p (new > usual): 0.0175
 - One-sided p (new < usual): 0.9825
- For complex systems, many scientists can justify either $\leq$ or $\geq$
 - Can “fudge” a one-sided test by picking the side after seeing the data
 - and getting $1/2$ the two-sided p-value
- So, two-sided preferred in most areas of science
- Even when only one side is scientifically relevant
 - e.g., Does spraying with fungicide increase yield?
 - e.g., Does a new teaching style increase understanding?
- Some scientific fields are more accepting of one-sided tests
 - e.g., toxicology

Why do you see t-tests instead of randomization tests?

Why does applied statistics usually use t-tests (model-based)

- Early days: hard to compute the sampling distribution
 - No longer a concern
- Design-based tests can be hard to implement
- Easy when the null hypothesis is simple
 - means are equal or regression slope = 0
- Hard for more complicated models
 - e.g. one specific regression coefficient in a multiple regression
 - An area of active statistical research
- Hard (but not impossible) to compute a design-based confidence interval
 - Current trend favors confidence intervals, not tests and p-values

Why randomization tests can be very useful

When a randomization test can save the day

- Force you to define a test statistic and a sampling scheme
- Allow more flexible null hypotheses, e.g.
 - ratio of means
 - difference of medians

Important Points

Important Points

- Randomization tests only assume treatments are randomly assigned
 - Design-based inference
- Null hypothesis more general than equal means
- Distinction between two-sided and one-sided hypotheses
 - And why common scientific practice is two-sided tests
- Why you generally see t-tests instead of randomization tests
 - Hard to implement for focused hypotheses